

Article

Colletotrichum fructicola Causes Necrotic Leaf Lesions in Avocado (*Persea americana*) in Amazonas, Peru: First Record and In Vitro Control Using *Piper* Essential Oils

Marly Guelac-Santillan ^{1,2}, Jherson Rojas-Vargas ¹, Elmer Chávez-Chacón ¹, Eryka Gaslac-Zumaeta ¹,
Manuel Oliva-Cruz ^{1,3,*} and Angel F. Huaman-Pilco ^{1,2,*}

¹ Grupo de Investigación en Patología Intracelular de Plantas, Instituto de Investigación para el Desarrollo Sustentable de Ceja de Selva, Universidad Nacional Toribio Rodríguez de Mendoza de Amazonas, Chachapoyas 01001, Amazonas, Peru; marly.guelac@untrm.edu.pe (M.G.-S.); rojasvargasjherson6@gmail.com (J.R.-V.); 7455977721@untrm.edu.pe (E.C.-C.); eryka.gaslac@untrm.edu.pe (E.G.-Z.)

² Grupo de Investigación en Raíces y Tuberosas (GIRT), Centro de Investigación en Germoplasma Vegetal y Mejoramiento Genético de Plantas (CIGEMP), Universidad Nacional Toribio Rodríguez de Mendoza de Amazonas, Chachapoyas 01001, Amazonas, Peru

³ Facultad de Ingeniería y Ciencias Agrarias, Universidad Nacional Toribio Rodríguez de Mendoza de Amazonas, Chachapoyas 01001, Amazonas, Peru

* Correspondence: manuel.oliva@untrm.edu.pe (M.O.-C.); angel.huaman@untrm.edu.pe (A.F.H.-P.)

Abstract

Anthraxnose is one of the most destructive diseases of avocado worldwide; however, foliar infections remain poorly documented in Peru. We investigated the etiology of necrotic leaf lesions observed in avocado plantations in Amazonas and assessed the in vitro antifungal activity of two *Piper* essential oils (EOs). Incidence was quantified in a 420-tree plot ($n = 150$ plants). Two representative isolates (MGS03, MGS04) were obtained from symptomatic leaves and characterized by morphology and multilocus sequencing (ITS, CHS-1, ACT, and TUB2). Maximum-likelihood phylogeny within the *C. gloeosporioides* species complex clustered with *C. fructicola*. Pathogenicity tests on detached leaves (unwounded inoculation) reproduced field symptoms and fulfilled Koch's postulates. The antifungal activity of *Piper carpunya* and *P. aduncum* EOs ($50\text{--}1000\ \mu\text{L L}^{-1}$) was evaluated using poisoned-medium assays, revealing a strong dose-dependent inhibition of mycelial growth. Inhibition reached $>89.8\%$ at $500\ \mu\text{L L}^{-1}$ and $\sim 100\%$ at $1000\ \mu\text{L L}^{-1}$, with *P. aduncum* showing slightly higher activity. This study provides the first confirmed record of *C. fructicola* associated with necrotic leaf lesions in avocado in Peru and demonstrates the promising antifungal potential of *Piper* EOs as eco-friendly candidates for anthracnose management.

Keywords: avocado; *Colletotrichum fructicola*; essential oils; fungal pathogens



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1. Introduction

Avocado (*Persea americana* Mill.) represents a crop of major economic and social relevance, driven by its high nutritional value and the steadily increasing demand across international markets [1]. Originating from Mexico and Central America, the avocado has been cultivated and consumed for over 8000 years [2]. Over the last two years, global avocado production has surpassed 10 million tons, an increase of over 10% compared with previous years [3,4]. Mexico continues to be the top exporter, followed by Peru, where avocado constitutes a strategic opportunity for local growers. Peru holds a prominent

position among the world's leading exporters, and its production represents a key pillar of the national economy [5]. In this context, in 2024, avocado exports exceeded 500 tons, and although the Amazonas region allocates a relatively small area to this crop, in 2025 it accounted for 27.6% of national production [6].

Despite its importance, the sustainability of avocado cultivation is threatened by several plant diseases. Among them, anthracnose stands out, caused by species of the genus *Colletotrichum* (Sordariomycetes, Ascomycota), which comprises a broad array of plant pathogens affecting vegetables, legumes, cereals, and numerous tropical and temperate crops worldwide [7]. This genus comprises major plant pathogens that cause significant economic losses across diverse agricultural systems globally [8,9]. In fruit trees and vegetables, *Colletotrichum* has been widely reported as the causal agent of infections in various plant organs, primarily in fruits [10,11]. Anthracnose, caused by *Colletotrichum* species, is widely distributed across avocado-producing regions and is extensively reported on both fruits and leaves throughout the crop cycle [12]. To the best of our knowledge, foliar infections of avocado caused by *Colletotrichum fructicola* have been reported in other avocado-producing countries; however, in Peru, previous studies have mainly documented fruit infections caused by *Colletotrichum*. Molecularly confirmed reports of foliar infections of avocado caused by *C. fructicola* have not yet been documented [2].

Colletotrichum comprises filamentous plant-pathogenic fungi with hundreds of species and substantial ecological diversity [13]. In avocado, the most frequently reported species belong to the *gloeosporioides* complex (*C. gloeosporioides*, *C. siamense*; [10,14]) and the *boninense* complex (*C. karstii*, *C. boninense*; [15]), along with members of the *acutatum* complex (*C. nymphaeae*; [16,17]) and *C. fructicola* [18]. Regionally, a high diversity associated with avocado has been documented: up to eleven species in Mexico [8,12], and multiple species confirmed as causal agents of anthracnose in Colombia and Brazil, particularly post harvest [14,17,19]. These patterns underscore the need for multilocus identification for reliable diagnosis of *Colletotrichum* on avocado [10,16].

Anthracnose in avocado is among the most widespread and severe diseases, affecting stems, leaves, flowers, and fruits, especially under high-humidity conditions [13]. Its management largely relies on fungicides; however, repeated applications promote resistance and generate residues that constrain market access [12]. Seeking sustainable alternatives, natural compounds have been evaluated; for instance, oregano essential oil achieved complete in vitro inhibition of *C. fructicola* in 'Hass' avocado via active packaging [20].

Plant-derived essential oils have emerged as promising disease control agents due to their broad-spectrum antifungal activity, biodegradability, and low toxicity [21,22]. Within this context, species of the genus *Piper* (Piperaceae) have gained increasing attention because of their strong inhibitory effects against phytopathogenic fungi. *Piper aduncum* and *Piper carpunya*, native and abundant in the Amazonas department, Peru, are rich in bioactive secondary metabolites such as phenylpropanoids, monoterpenes, and sesquiterpenes, including dillapiole, safrole, and 1,8-cineole, which have been reported to disrupt fungal cell membranes, inhibit spore germination, and suppress mycelial growth [23–25].

Essential oil of *P. aduncum* has demonstrated significant in vitro antifungal activity against several plant pathogens, including *Colletotrichum* spp., *Fusarium* spp., and *Botrytis cinerea*, achieving high levels of mycelial growth inhibition and reduced conidial viability [25,26]. Similarly, *P. carpunya* essential oil has been shown to inhibit phytopathogenic fungi through alterations in membrane permeability and interference with key metabolic pathways [25,27,28]. The documented antifungal efficacy of these essential oils, together with their natural availability in the Peruvian Amazon and their low environmental impact, supports their selection as sustainable alternatives for the management of avocado anthracnose.

In avocado orchards in the Amazonas department of Peru, we observed leaves with necrotic lesions suggestive of anthracnose. This study aimed to (i) identify the causal agent through morphological and multilocus characterization, (ii) assess its foliar pathogenicity, and (iii) evaluate the in vitro antifungal activity of *Piper* essential oils. To our knowledge, this is the first confirmed report of *Colletotrichum fructicola* associated with foliar necrosis in avocado in Peru.

2. Materials and Methods

2.1. Sampling, Incidence, and Isolation

Avocado leaves with necrotic lesions were sampled from the UNTRM Experimental Station in Magdalena, Chachapoyas, Amazonas, Peru (6.400251 S 77.885862 W; 1939 m a.s.l.). Disease incidence was assessed in 150 trees within a 420-tree plot by recording the presence or absence of necrotic leaf lesions per plant, and expressed as the percentage of symptomatic trees. Affected leaves typically showed irregular dark brown spots starting at the margins or apex and expanding along the lamina (Figure 1).



Figure 1. Field symptoms at the UNTRM Experimental Station, Magdalena, Chachapoyas (Amazonas, Peru). (A,B) Necrotic leaf lesions.

For fungal isolation, symptomatic leaves were washed, surface-disinfected with 3% NaClO for 2 min, rinsed twice in sterile distilled water, and air-dried under sterile conditions. Approximately 1 mm fragments from lesion margins were placed on Potato Dextrose Agar (PDA) and incubated at 25 °C for 7 days [15]. Colonies with morphology consistent with *Colletotrichum* were transferred to fresh PDA and purified by single-spore isolation. Two representative isolates (MGS03 and MGS04) were selected for morphological and molecular characterization, pathogenicity tests, and in vitro essential oil assays.

2.2. Morphological Identification

Isolates MGS03 and MGS04 were grown on PDA plates and incubated at 25 ± 2 °C. Colony diameter was measured after 7 days along two perpendicular axes, and the radial growth rate (mm day^{−1}) was calculated [29]. Cultural and morphological traits were assessed following the methods of Damm [30]. Micromorphology was examined using the slide culture technique [31]. Small agar blocks from the colony margin were placed

on PDA-coated glass slides and incubated in moist chambers at 25 °C until sporulation. Permanent mounts were prepared in lactophenol cotton blue, and conidiophores, conidiogenous cells, conidia, and appressoria were observed with an Olympus microscope equipped with a DP74 digital camera (Olympus, Tokyo, Japan). For each isolate, at least 50 measurements per structure were taken to determine size ranges and mean $\pm$ standard deviation. Morphological features were compared with published descriptions of species within the *Colletotrichum gloeosporioides* complex [30].

2.3. Molecular Analysis and Phylogeny

Total DNA was extracted using the Wizard® Genomic DNA Purification Kit (Promega, WI, USA) according to the manufacturer's instructions. DNA concentration was quantified using a BioSpectrometer® basic (Eppendorf, NJ, USA) and subsequently diluted to 0.5 ng/ μ L for PCR reactions. Four informative regions were amplified: the internal transcribed spacer region (ITS) (ITS1, 5.8S, and ITS2), chitin synthase (CHS-1), partial actin (ACT), β -tubulin (TUB2). The primer pair ITS-1F/ITS4R was used for amplification [32] CHS-79F/CHS-345R, ACT-512F/ACT-783R [33], BT2a/BT2b [34]. The amplified PCR products were purified and sequenced using the Sanger method. Sequences were edited with Sequencher version 5.4.6 (Gene Codes, Ann Arbor, MI, USA) [35]. Edited sequences were aligned with MUSCLE [36] implemented in MEGA 11.0 [37]. Subsequently, they were concatenated with Seaview [38]. The phylogenetic analysis by maximum likelihood was performed on the CIPRES Science Gateway platform [39]. The resulting tree was midpoint-rooted and edited using iTools version 4.5.1.9. [40]. GenBank accession numbers of the sequences used in the phylogenetic analysis are listed in Table 1.

Table 1. GenBank accession numbers of DNA sequences of *Colletotrichum gloeosporioides* species complex included in phylogenetic study.

Species	Culture	Host	Country	Gen Bank Accession Number			
				ITS	ACT	CHS	TUB2
<i>Colletotrichum theobromicola</i>	ICMP 18649	<i>Theobroma cacao</i>	Panama	JX010294	X009444	JX009869	JX010447
<i>Colletotrichum ti</i>	ICMP 4832	<i>Cordyline</i> sp.	New Zealand	JX010269	JX009520	JX009898	JX010447
<i>Colletotrichum tropicale</i>	ICMP 18653	<i>Theobroma cacao</i>	Panama	JX010264	JX009489	JX009870	JX010412
<i>Colletotrichum alienum</i>	ICMP 12071	<i>Malus domestica</i>	New Zealand	JX010251	JX009572	JX009882	JX010416
<i>Colletotrichum xanthorrhoea</i>	ICMP 17903	<i>Xanthorrhoea preissii</i>	Australia	JX010261	JX009478	JX009823	JX010453
<i>Colletotrichum aenigma</i>	ICMP 18608	<i>Persea americana</i>	Israel	JX010244	JX009443	JX009774	JX010394
<i>Colletotrichum queenslandicum</i>	ICMP 1778	<i>Carica papaya</i>	Australia	JX010276	JX009447	JX009899	JX010419
<i>Colletotrichum asianum</i>	ICMP 18580	<i>Coffea arabica</i>	Thailand	FJ972612	JX009584	JX009867	JX010411
<i>Colletotrichum psidii</i>	ICMP 19120	<i>Psidium</i> sp.	Italy	JX010219	JX009515	JX009901	JX010448
<i>Colletotrichum clidemiae</i>	ICMP 18658	<i>Clidemia hirta</i>	USA	JX010265	JX009537	JX009877	JX010443
<i>Colletotrichum salsolae</i>	ICMP 19051	<i>Salsola tragus</i>	Hungary	JX010242	JX009562	JX009863	JX010408
<i>Colletotrichum siamense</i>	ICMP 18578	<i>Coffea arabica</i>	Thailand	JX010171	FJ907423	JX009865	JX010409
<i>Colletotrichum fructicola</i>	ICMP 18581	<i>Coffea arabica</i>	Thailand	JX010165	FJ907426	JX009866	JX010410
<i>Colletotrichum fructicola</i>	ICMP 12568	<i>Persea americana</i>	Australia	JX010166	JX009529	JX009762	-
<i>Colletotrichum fructicola</i>	MGS04	<i>Persea americana</i>	Peru	PX518095	PX664117	PX664115	PX664119
<i>Colletotrichum fructicola</i>	MGS03	<i>Persea americana</i>	Peru	PX518094	PX664118	PX664116	PX664120

Table 1. Cont.

Species	Culture	Host	Country	Gen Bank Accession Number			
				ITS	ACT	CHS	TUB2
<i>Colletotrichum nupharicola</i>	ICMP 18187	<i>Nuphar lutea</i> subsp. <i>polysepala</i>	USA	JX010187	JX009437	JX009835	JX010403
<i>Colletotrichum musae</i>	ICMP 19119	<i>Musa</i> sp.	USA	JX010146	JX009433	JX009896	HQ596285
<i>Colletotrichum aotearoa</i>	ICMP 18537	<i>Coprosma</i> sp.	New Zealand	JX010205	JX009564	JX009853	JX010425
<i>Colletotrichum horii</i>	ICMP 10492	<i>Diospyros kaki</i>	Japan	GQ329690	JX009438	JX009752	JX010455
<i>Colletotrichum perseae</i>	GA272	<i>Persea americana</i>	Israel	KX620321	KX620158	-	KX620354
<i>Colletotrichum perseae</i>	CBS 141365	<i>Persea americana</i>	Israel	KX620308	ARE68452	-	WLV27715
<i>Colletotrichum alatae</i>	ICMP 17919	<i>Dioscorea alata</i>	India	JX010190	JX009471	JX009837	JX010388
<i>Colletotrichum alatae</i>	ICMP 18122	<i>Dioscorea alata</i>	Nigeria	JX010191	JX009470	JX009846	JX010449
<i>Colletotrichum aeschynomenes</i>	ICMP 17673	<i>Aeschynomene virginica</i>	USA	JX010176	JX009483	JX009799	JX010397
<i>Colletotrichum gloeosporioides</i>	ICMP 17821	<i>Citrus sinensis</i>	Italy	JX010152	JX009531	JX009818	JX010450
<i>Colletotrichum gloeosporioides</i>	ICMP 18738	<i>Carya illinoensis</i>	Australia	JX010151	JX009542	JX009797	-

2.4. Pathogenicity Tests

Pathogenicity tests were performed on young avocado (*Persea americana*) leaves under controlled laboratory conditions. Leaves were surface-sterilized following the protocol described by [30] and placed in moist chambers. Two inoculation methods were used. In the first, mycelial disks (5 mm diameter) from actively growing cultures were placed on slightly wounded leaf tissues. In the second, a conidial suspension (1×10^6 conidia mL⁻¹) was applied to unwounded leaf surfaces to simulate natural infection. Sterile distilled water was used as a negative control. Each isolate was tested with six replicates per method, including controls. Inoculated leaves were incubated at 25 °C under high humidity and monitored daily for symptom development. To fulfill Koch's postulates, the pathogen was reisolated from symptomatic tissues and cultured on potato dextrose agar (PDA). Conidial masses were documented using a stereomicroscope (SMZ18, Nikon, Tokyo, Japan), and detailed observations were performed by scanning electron microscopy (SEM; ZEISS Smart SEM v7.05, Oberkochen, Germany) after drying infected tissues in silica gel for 24 h.

2.5. In Vitro Inhibition Test with Essential Oils

The antifungal activity of essential oils from *Piper aduncum* and *Piper carpunya*, provided by the Plant Health Research Laboratory, was evaluated against *C. fructicola*. The assay was conducted using the poisoned food technique [41]. Five different concentrations of both oils (50, 150, 250, 500, and 1000 µL L⁻¹) were prepared, dissolved in absolute ethyl alcohol (1:2 v/v), and incorporated into molten PDA (Potato Dextrose Agar) cooled to 45 °C. Twenty milliliters (20 mL) of the solution from each concentration were added to sterile Petri dishes and allowed to solidify. Subsequently, 6 mm diameter agar disks containing seven-day-old *C. fructicola* mycelia from the active growth zones were cut and placed in the center of each plate. The assays included a control treatment (without oils). Additionally, two reference controls were included: (i) an absolute ethanol control (EtOH, ≥99.8%) at the same solvent proportion used for essential oil dilution, and (ii) a commercial fungicide control at the recommended dose. Five plates per treatment were used and incubated at

24 °C. Colony diameters were measured with a digital vernier caliper every 48 h for ten days. The percentage of Mycelial Growth Inhibition (MGI) was calculated as follows:

$$\text{MGI (\%)} = \left(\frac{D_c - D_t}{D_c} \right) \times 100$$

where D_c is the mean colony diameter in the control, and D_t is the mean colony diameter in the essential oil treatment.

2.6. Statistical Analysis

The relationship between essential oil concentration and mycelial growth inhibition (MGI) was modeled using a nonlinear log–logistic regression implemented in the *drc* package in R to estimate ED_{50} and ED_{90} values. The model was fitted as follows:

$$f(x) = \frac{100}{1 + \exp[b(\log(x) - \log(ED_{50}))]}$$

where b is the slope of the dose–response curve and ED_{50} represents the concentration producing 50% inhibition. Lower and upper limits were fixed at 0 and 100%, respectively.

3. Results

3.1. Incidence

In the UNTRM avocado plot, necrotic leaf lesions caused by *Colletotrichum fructicola* were detected in 60% of the 150 evaluated trees. Affected leaves developed irregular dark brown to black necrotic lesions that typically originated at the margins or apex and expanded toward the lamina center, often coalescing into larger necrotic areas (Figure 1). In advanced stages, severe necrosis led to partial defoliation. Under high humidity, pink to orange conidial masses characteristic of *Colletotrichum* were observed on the lesion surface.

3.2. Morphological and Molecular Characterization

In this study, two isolates obtained from symptomatic leaves (MGS03 and MGS04) displayed cultural and morphological traits consistent with the *C. fructicola* species complex as described by Weir et al. [10].

Colletotrichum fructicola Prihastuti L. Cai & K.D. Hyde, Fungal Diversity 39: 158. 2009

Macroscopic characteristics (Figure 2A): After seven days on PDA at 25 ± 2 °C, *C. fructicola* showed an average growth rate of 29.6 ± 0.19 mm/day, with dense, velvety mycelium and progressively dark pigmentation. The colonies displayed concentric zones ranging from gray to black, with the reverse showing characteristic brown to black pigmentation. Microscopic characteristics (Figure 2B–D): Conidiophores were hyaline, septate, and cylindrical. Conidia were hyaline, smooth, and cylindrical to clavate in shape, measuring on average 16.00×4.04 µm (range: $15.08\text{--}17.08 \times 3.63\text{--}4.39$ µm; $n = 50$). Appressoria were brown, solitary or grouped, elliptic to lobed in outline, and developed from vegetative hyphae.

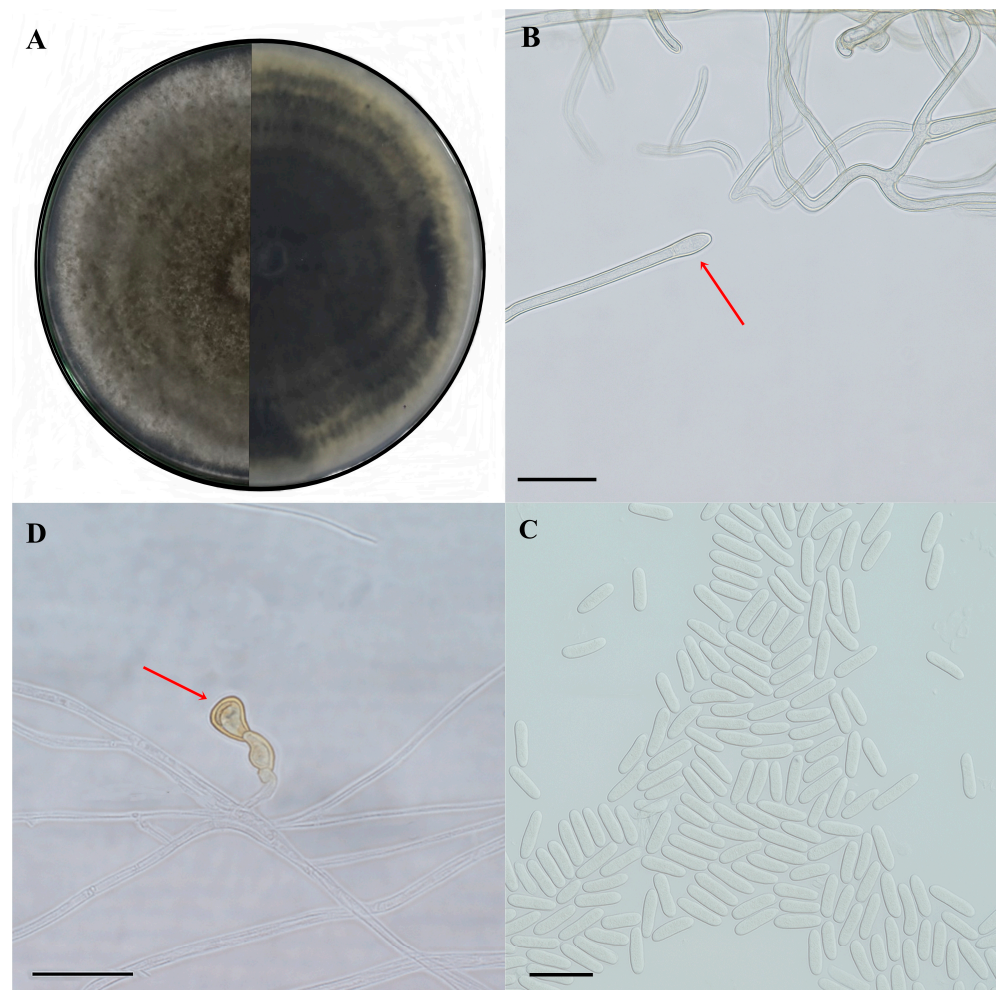


Figure 2. *C. fructicola*. (A) Upper and lower views of colonies grown on potato dextrose agar (PDA) and incubated at 28 °C for seven days; (B) conidiophores (red arrow); (C) conidia; (D) appressoria (red arrow). Scale bars: 1 mm (A), 20 μm (B,D), 50 μm (C).

Molecular analysis: From the two isolates obtained, *C. fructicola* MGS03 and MGS04, sequences were generated for each. Alignment of the ITS, CHS-1, ACT and TUB2 markers produced a dataset comprising 1863 characters, including gaps: ITS (1–592), CHS-1 (593–865), ACT (866–1131), and TUB2 (1132–1863). Phylogenetic analysis was conducted under the Transitional model with equal base frequencies and a discrete Gamma model to account for rate heterogeneity, as estimated by jModelTest. The phylogenetic analysis based on the concatenated multilocus dataset [42] revealed that isolates MGS03 and MGS04 clustered with the type strain *Colletotrichum fructicola* ICMP 18581, with 93% bootstrap support (Figure 3). Notes: *C. fructicola* was initially reported in coffee berries as an endophyte and pathogen in Thailand [43] and later as an endophyte in cocoa leaves (*Theobroma cacao*), corresponding to the ex-holotype crop of *C. ignotum* (syn. *C. fructicola*; CBS 125397 = ICMP 18646) [44,45]. In 2022, the species was isolated in China from avocado fruits (*Persea americana* cv. Hass) with symptoms of anthracnose, exported from Peru [46]. In 2024, *C. fructicola* was isolated from anthracnose cocoa leaves in the Amazonas region, Peru [47]. In this study, *C. fructicola* is reported for the first time causing foliar lesions in avocado in Peru.

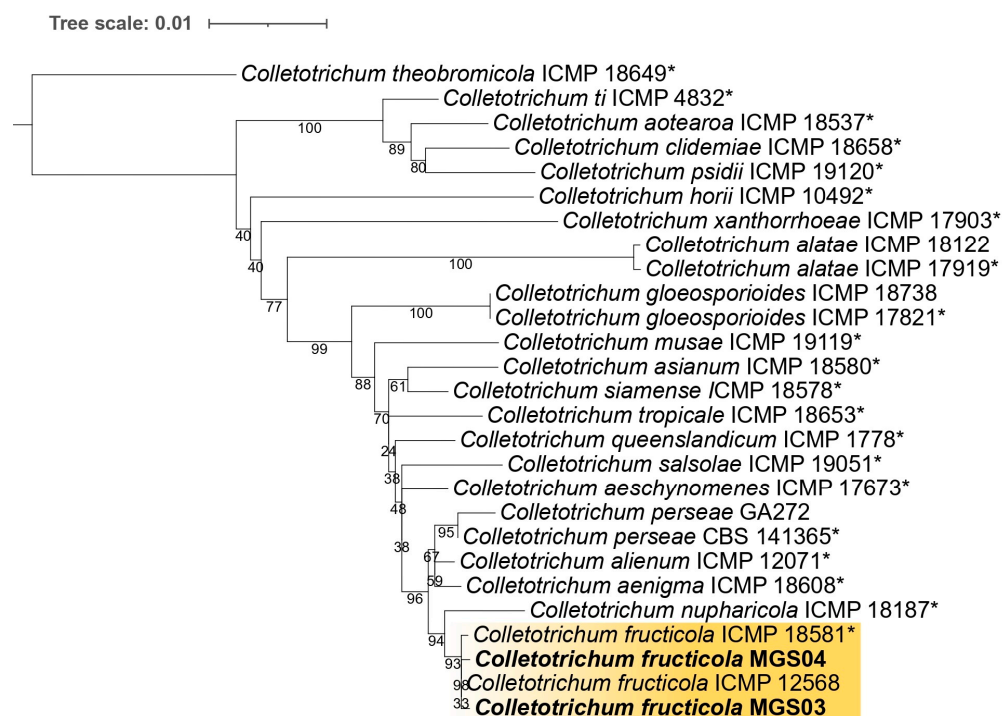


Figure 3. Phylogenetic analysis of isolates belonging to the *C. fruticola* species complex included isolates obtained in this study. A concatenated dataset of (ITS, ACT, CHS-1, and TUB2) was used to construct the phylogeny. Isolates obtained in this study are shown in bold. The yellow shading indicates the *C. fruticola* clade. Asterisks (*) denote type or reference sequences.

3.3. Pathogenicity Tests

Pathogenicity assays on detached avocado leaves confirmed that isolates MGS03 and MGS04 reproduced the disease symptoms observed in the field. Three days after inoculation, inoculated leaves developed typical anthracnose lesions, characterized by dark necrotic spots expanding from the inoculation point to the mycelial disk (Figure 4B) and conidial suspension (Figure 4C). No symptoms were recorded in non-inoculated control leaves (Figure 4A). Conidiomata and conidial masses formed on inoculated tissues (Figure 4D), and fungal colonization was detailed by SEM (Figure 4E). The pathogen was successfully reisolated from symptomatic tissues and identified as *C. fruticola* based on its cultural and morphological features, thereby fulfilling Koch's postulates.

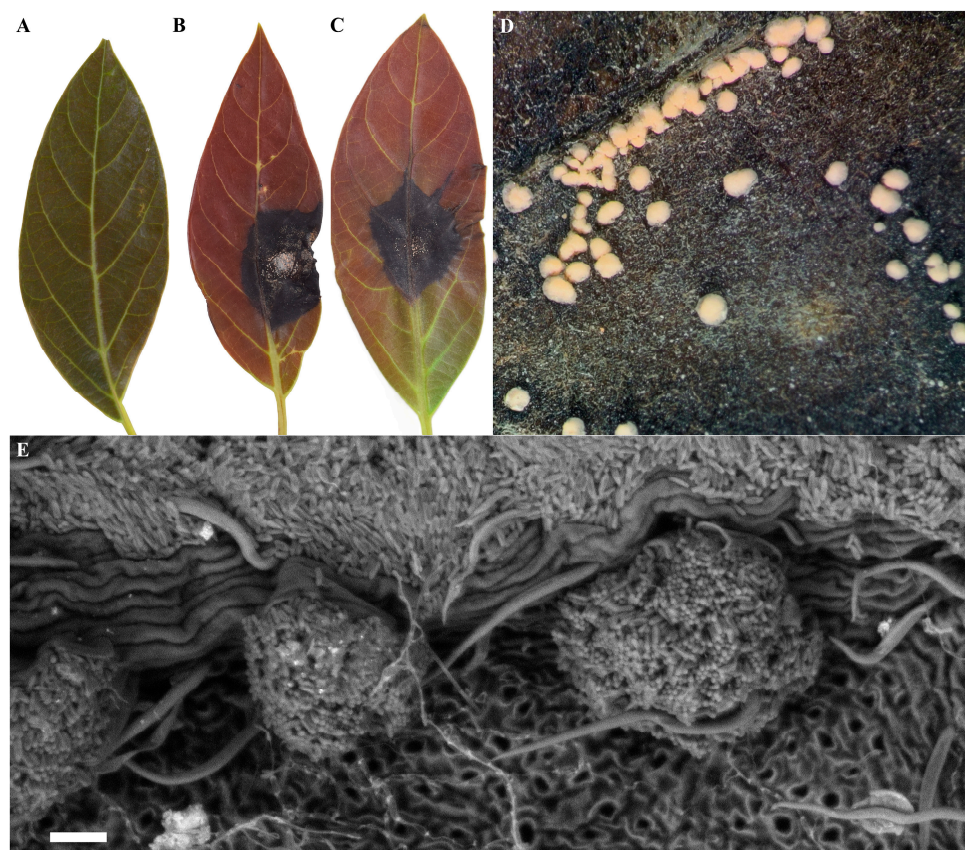


Figure 4. Pathogenicity test of *C. fructicola*. (A) Non-inoculated leaf; (B) leaves inoculated with mycelial disks; (C) leaves inoculated with conidial suspension; (D) presence of reproductive structures (conidial masses); (E) conidial masses observed by SEM, Scale bar = 30 μm .

3.4. Inhibition Test

The *in vitro* inhibition assay revealed that both *P. aduncum* and *P. carpunya* essential oils (EOs) significantly inhibited the mycelial growth of *C. fructicola* in a clear dose-dependent manner (Table 2; Figure 5). At lower concentrations (50–150 $\mu\text{L L}^{-1}$), inhibition was moderate and remained below 30% for both EOs. At 250 $\mu\text{L L}^{-1}$, *P. aduncum* exhibited a markedly stronger inhibitory effect (58.2%) than *P. carpunya* (23.9%). A sharp increase in antifungal activity was observed from 500 $\mu\text{L L}^{-1}$ onward, with *P. aduncum* achieving inhibition values close to 90%, while *P. carpunya* reached approximately 57%. At the highest concentration (1000 $\mu\text{L L}^{-1}$), both essential oils completely inhibited fungal growth, reaching 100% inhibition. Overall, *P. aduncum* consistently showed higher antifungal potency than *P. carpunya* across all tested concentrations.

Table 2. Mycelial growth and inhibition (%) of *C. fructicola* under different concentrations of essential oils in the *in vitro* inhibition assay.

Treatment	Concentration ($\mu\text{L L}^{-1}$)	Mycelial Growth (mm)		Inhibition (%)	
		<i>P. aduncum</i>	<i>P. carpunya</i>	<i>P. aduncum</i>	<i>P. carpunya</i>
Control	0	70.34 $\pm$ 6.40 ^a	70.34 $\pm$ 7.93 ^a	0.0	0.0
Control fungicide	3.75 g L ⁻¹	13.74 $\pm$ 0.98 ^f	13.74 $\pm$ 0.97 ^f	80.5	80.5
Dose 1	50	57.66 $\pm$ 10.29 ^b	65.44 $\pm$ 7.33 ^b	18.0	7.0
Dose 2	150	50.20 $\pm$ 7.59 ^c	58.34 $\pm$ 9.48 ^c	28.6	17.1
Dose 3	250	29.42 $\pm$ 4.06 ^d	53.52 $\pm$ 6.47 ^d	58.2	23.9
Dose 4	500	7.14 $\pm$ 5.60 ^e	30.50 $\pm$ 3.56 ^e	89.8	56.6
Dose 5	1000	0.00 $\pm$ 0.00 ^f	0.00 $\pm$ 0.00 ^f	100	100

Values are means $\pm$ SD (n = 5). Different letters within a column indicate significant differences according to Tukey's test ($p \leq 0.05$). Inhibition percentages were calculated relative to the untreated control.

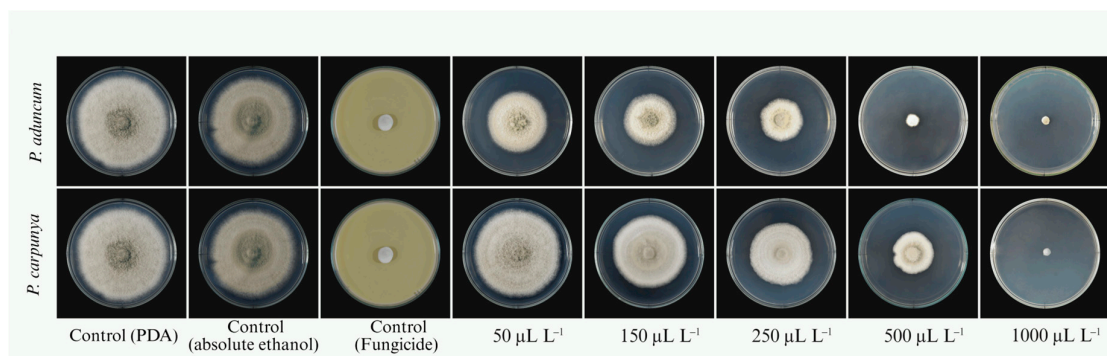


Figure 5. In vitro inhibition testing of *C. fructicola*. Marked inhibition of mycelial growth is observed in the presence of essential oils compared with the controls.

This dose-dependent trend was further supported by dose–response modeling, which showed lower effective dose values for *P. aduncum* ($ED_{50} = 209.37 \pm 13.95 \mu\text{L L}^{-1}$; $ED_{90} = 604.40 \pm 101.00 \mu\text{L L}^{-1}$; $R^2 = 0.93$) compared with *P. carpunya* ($ED_{50} = 385.60 \pm 24.22 \mu\text{L L}^{-1}$; $ED_{90} = 1114.18 \pm 150.12 \mu\text{L L}^{-1}$; $R^2 = 0.92$), confirming the higher antifungal efficacy of *P. aduncum* against *C. fructicola* (Figure 6).

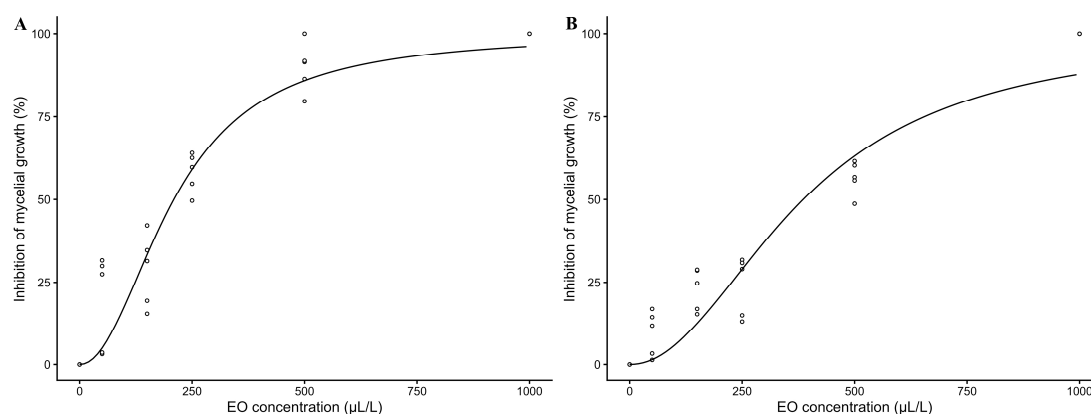


Figure 6. Effects of different concentrations of essential oils from *Piper aduncum* (A) and *P. carpunya* (B) on mycelial growth of *Colletotrichum fructicola*. Dose–response curves fitted using logistic model.

4. Discussion

In this study, *Colletotrichum fructicola* was confirmed as the causal agent of necrotic leaf lesions in avocado in the Amazonas region of Peru. The symptomatology observed in the field—irregular dark brown to black spots originating at the leaf margins, frequently coalescing into larger necrotic patches—is consistent with descriptions of foliar anthracnose in avocado and other tropical hosts [13,36]. The recorded incidence (60%) reflects a high disease pressure under local environmental conditions, particularly the region’s high humidity, which favors sporulation, secondary infections, and the rapid expansion of necrotic lesions [37]. High foliar damage of this type is known to reduce photosynthetically active leaf area and compromise plant vigor and productivity [38,39].

The high disease incidence recorded in this study (60%) indicates intense disease pressure and can be largely explained by the prevailing agroclimatic conditions of the Amazonas region. This area is characterized by persistently high relative humidity, frequent rainfall events, and moderate temperatures, all of which are known to favor *Colletotrichum* infection processes, including conidial germination, appressorium formation, and secondary dispersal [11,16]. Prolonged leaf wetness periods enhance the establishment and expansion of necrotic lesions, accelerating epidemic development in perennial crops such

as avocado [48]. Severe foliar damage of this nature has been shown to significantly reduce the photosynthetically active leaf area, negatively affecting carbon assimilation, plant vigor, and ultimately yield potential [49,50].

Previous studies in Peru have primarily reported anthracnose in avocado fruits rather than leaves, with *C. siamense* and *C. fruticola* identified as predominant causal agents in postharvest and field infections [51]. These studies documented lower disease incidence or focused on fruit symptomatology, reflecting differences in host tissue susceptibility, environmental exposure, and management practices. In contrast, the present work demonstrates that *C. fruticola* is capable of causing severe foliar infections under the humid conditions of the Amazonas region, suggesting a broader epidemiological role for this species in Peruvian avocado systems.

Comparatively, *C. siamense* has been reported as highly aggressive in fruit infections under warm and humid conditions, while *C. fruticola* has been associated with both fruit and leaf symptoms in other avocado-producing countries [10,52]. However, molecularly confirmed reports of foliar anthracnose caused by *C. fruticola* in avocado had not been previously documented in Peru. Therefore, this study represents the first molecular evidence of *C. fruticola* infecting avocado leaves in the country, expanding current knowledge on the diversity, tissue specificity, and epidemiological relevance of *Colletotrichum* species affecting avocado. Overall, the high incidence observed highlights the need to consider foliar anthracnose as an emerging constraint in humid avocado-growing regions of Peru. Integrated disease management strategies that incorporate climatic risk, early detection, and targeted control measures are essential to mitigate the impact of *Colletotrichum*-induced foliar damage and sustain avocado productivity under increasingly favorable conditions for pathogen development.

The morphological and cultural traits of isolates MGS03 and MGS04 matched those described for members of the *C. gloeosporioides* species complex, characterized by dense, dark colonies with concentric zonation and cylindrical hyaline conidia [10,11]. Multilocus phylogenetic analysis (ITS, CHS-1, ACT, and TUB2) placed both isolates in a well-supported clade with the type strain *Colletotrichum fruticola* ICMP 18581, confirming their taxonomic identity. This multilocus approach is essential for species delimitation within *Colletotrichum*, a genus known for its cryptic diversity and morphological overlap among species complexes [36,44]. Comparable diversity patterns have been reported in avocado-producing countries such as Mexico, Brazil, Israel, and Colombia, where several phylogenetically distinct *Colletotrichum* species co-occur and may infect different plant organs [8,12,14,17,19]. The present work expands the geographical distribution of *C. fruticola* affecting avocado leaves and constitutes the first confirmed report for Peru.

Pathogenicity tests reproduced the characteristic necrotic lesions within three days post inoculation, fulfilling Koch's postulates and confirming the foliar pathogenicity of the isolates. Similar rapid lesion development has been documented in pathogenicity assays of *Colletotrichum* spp. on avocado and mango leaves [37,45]. The rapid onset of symptoms observed here suggests a high virulence potential, particularly under conditions favorable to fungal growth. Previous studies indicate that species within the *C. gloeosporioides* complex exhibit considerable variability in aggressiveness and tissue specificity [42,43], which may explain the severe foliar deterioration observed in the field.

The essential oils of *P. aduncum* and *P. carpunya* displayed strong, dose-dependent antifungal activity against *C. fruticola*, with inhibition reaching 89.8% at 500 $\mu\text{L L}^{-1}$ for *P. aduncum* and 56.6% for *P. carpunya*, and both oils fully inhibited mycelial growth (100%) at 1000 $\mu\text{L L}^{-1}$. These results agree with previous studies reporting potent activity of *Piper* essential oils against phytopathogenic fungi, including *Moniliophthora roreri*, *Alternaria alternata*, and other *Colletotrichum* species [46,47]. Their antifungal mechanism

involves phenylpropanoids and terpenoids—such as dillapiole, safrole, eugenol, and β -caryophyllene—which disrupt membrane integrity, increase permeability, and interfere with essential enzymatic processes [48]. The high sensitivity of *C. fruticola* to *Piper* metabolites underscores the potential of these oils as natural antifungal agents, particularly in regions where synthetic fungicide use is restricted or resistance has been reported.

Overall, this study provides new evidence of the epidemiological relevance of *C. fruticola* as a foliar pathogen in Peruvian avocado orchards and demonstrates the strong inhibitory effect of two *Piper* essential oils against this pathogen. These findings highlight the importance of integrating molecular diagnostics with sustainable disease management strategies and position the *Piper* genus as a promising source of bioactive compounds for anthracnose control in tropical agriculture. Future studies should evaluate the efficacy of *Piper* essential oils under greenhouse and field conditions, as well as their compatibility with integrated disease management programs.

5. Conclusions

This study reports *Colletotrichum fruticola* as the causal agent of necrotic leaf lesions in avocado in the Amazonas department, Peru. Additionally, the essential oils of *P. aduncum* and *P. carpunya* exhibited potent mycelial inhibition (100% at 1000 $\mu\text{L L}^{-1}$), underscoring their potential as effective natural alternatives to synthetic fungicides. Together, these results highlight the *Piper* genus as a promising source of sustainable bioactive compounds for the integrated management of avocado anthracnose.

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